Constrained-worldline brachistochrones in non-stationary spacetimes: Kodama rails, closed forms, and plunge inversion

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We extend the relativistic brachistochrone (fastest-descent constrained rail curves, Perlick 1991) from stationary to non-stationary spacetimes. The rail constraint $-u \cdot W = \hat{E}$ is an actively controlled invariant, legitimate via Pontryagin's principle; the selector W follows a hierarchy Killing $\to$ conformal Killing $\to$ Kodama vector. We treat FLRW (degenerate base), Vaidya (dynamical black hole, $W = \partial_v$ the Kodama vector), and Thakurta–Kerr (conformal rotating black hole). Main results: Kodama energy conservation along the rail; the plunge-radius law and the absence of a *physical* evaporative inversion (only rotation and the conformal factor invert); closed-form equatorial separatrices in Weierstrass functions with Doran continuation through the ergosphere; the trichotomy at the ergosphere with an analytic cusp theorem; and an existence theorem for the conformal plunge inversion.

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CONTENTS

I. INTRODUCTION

The relativistic brachistochrone—the fastest constrained "rail" worldline between two events—was formulated by Perlick for stationary spacetimes [1, 2], where a timelike Killing vector provides the conserved rail energy. It connects naturally to Fermat's principle and to optical (Randers–Zermelo–Finsler) geometry [3–7]. Building on our earlier relativistic-brachistochrone studies—the stationary Kerr geometry [8], the interior Schwarzschild metric [9], and Tolman–Oppenheimer–Volkoff stellar interiors [10]—we extend the construction to *non-stationary* spacetimes: FLRW, Vaidya [11], and Thakurta–Kerr [12]. These extensions raise three questions that organize the paper. *(i) Is the rail constraint legitimate without a Killing vector?* We show it is: $-u \cdot W = \hat{E}$ is an actively controlled invariant, and the fastest constrained worldline solves a well-posed time-optimal control problem (Sec. II) whose control cost measures precisely the missing symmetry. *(ii) What replaces the conserved energy?* In a spherically symmetric dynamical spacetime the selector W is the Kodama vector, whose flux charge is the Misner–Sharp mass; the rail conserves the Kodama energy identically even though it is *not* Noether-conserved for free fall (Sec. IV). *(iii) Which phenomena are genuinely non-stationary?* We isolate three: capture by cosmological expansion, the teleological (future-anticipating) turning point in Vaidya, and the breathing freezing surface that eats the radius of convergence of the quasi-constants in Thakurta–Kerr. The three spacetimes form a ladder of increasing structure—FLRW (degenerate base), Vaidya (dynamical horizon), Thakurta–Kerr (rotation + conformal expansion)—treated in Secs. III–V; equatorial closed forms and the plunge-inversion analysis follow in Secs. VI–VI D.

II. THE CONTROLLED-RAIL PRINCIPLE

A. Rail constraint and the indicatrix

The rail is a constrained worldline with the invariant

$$-u \cdot W \;=\; \hat{E}, \tag{1}$$

actively maintained by an ideal (magnetic-type) rail force $f_\mu = Du_\mu/\mathrm{d}\tau$ with $f \cdot u = 0$, so that the rail does no work on the particle.

a. The ellipse. Fix an event and a time function adapted to W. The linear constraint (1) fixes the W-component of u; the mass shell $u \cdot u = -1$ is a quadric; their intersection, projected onto the physical velocity plane $(\mathrm{d}r,\, r\,\mathrm{d}\varphi)$ per unit selector-time, is an ellipse

$$\dot{x}(\theta) = c + R(\theta), \qquad \theta \in [0, 2\pi), \tag{2}$$

whose center c is the Zermelo wind (zero for FLRW, ingoing radial for Vaidya, azimuthal—frame dragging—for Thakurta–Kerr) and whose axes R shrink with the local speed as the freezing surface is approached (Fig. 1). Every branch—arrival time t, proper time τ, conformal time η—shares this *same* indicatrix; the branches differ only in the cost one-form, i.e. in which clock measures the elapsed travel time.

b. Pontryagin reduction. Minimizing travel time subject to (2) is a time-optimal control problem with control θ. Since the admissible-velocity set is a smooth strictly convex oval, the maximizer of $p \cdot \dot{x}(\theta)$ is unique and Pontryagin's maximum principle [13, 14] delivers the branch Hamiltonian $H(x, p)$ in closed form (Secs. IV–V); the convex-oval regularity makes the vakonomic and d'Alembert formulations coincide, so no Filippov relaxation is needed. Two structural facts hold throughout: the transversality condition

$$H = 0 \qquad \text{(free arrival time)}, \tag{3}$$

and $p_\varphi = J$ conserved (axial symmetry). In the stationary limit H is autonomous and $H \equiv 0$ is a genuine first integral; non-stationarity enters only through the explicit time dependence $\mathrm{d}H/\mathrm{d(time)} = \partial_{\mathrm{time}} H$, proportional to $m'(v)$ (Vaidya) or $A'(\eta)$ (Thakurta–Kerr).

c. Cost of control. Differentiating the invariant along the worldline,

$$\frac{\mathrm{d}(-u \cdot W)}{\mathrm{d}\tau} = -\,u^a u^b\, \nabla_{(a} W_{b)}, \tag{4}$$

which vanishes for *all* u iff W satisfies the Killing equation. When it does not, the right-hand side is the power the rail force must supply to hold $\hat{E}$ fixed: the control cost equals the failure of the Killing equation, i.e. the missing symmetry. This is why the non-stationary rail is legitimate but not free—$\hat{E}$ is controlled, not conserved. The optical reductions of H reproduce the Randers/Zermelo structure [6, 15].

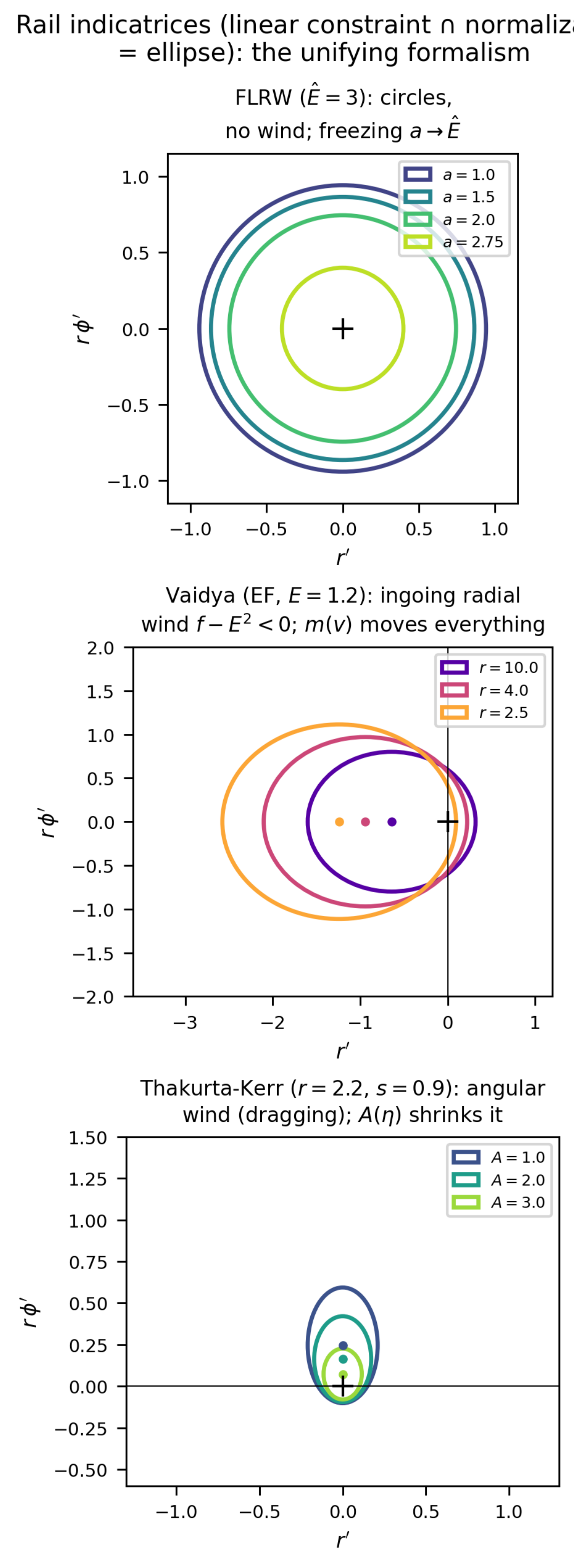

FIG. 1. Rail indicatrices (linear constraint ∩ normalization = ellipse): the unifying formalism. FLRW gives shrinking circles (no wind); Vaidya an ingoing radial wind ($f - E^2 < 0$, driven by $m(v)$); Thakurta–Kerr an angular wind (frame dragging), the ellipse shrinking with $A(\eta)$.

B. Selecting W: the Killing–CKV–Kodama hierarchy

The selector W is fixed by a hierarchy of decreasing symmetry, each level recovering the previous one in the appropriate limit.

(a) Killing. If a timelike Killing vector ξ exists, $W = \xi$ and the right-hand side of Eq. (4) vanishes identically: $-u \cdot \xi$ is Noether-conserved and the rail is free. This is Perlick's stationary case [1].

(b) Conformal Killing. If the spacetime is conformally stationary, $g = \Omega^2 \hat{g}$ with $\hat{g}$ stationary, the timelike conformal Killing vector ∂_η selects the conformal-time branch; $-u \cdot W$ is exactly conserved for null rays and controlled up to the conformal factor for massive rails. FLRW and constant-A Thakurta–Kerr live at this level.

(c) Kodama. In a spherically symmetric *dynamical* spacetime with no Killing or conformal Killing vector, the Kodama vector $K^a = \varepsilon^{ab}\nabla_b r$ [16, 17] is the canonical replacement: it is divergence-free (Eq. (8)), reduces to ξ in the static limit, and its flux $G^{ab}K_b$ is conserved with charge the Misner–Sharp mass. Vaidya is the paradigm (Sec. IV A).

(d) Convention. Absent even spherical symmetry the Kodama construction fails and W becomes a declared convention, with no distinguished normalization of the cost one-form. Levels (a)–(c) are what make the non-stationary rail canonical rather than arbitrary.

III. FLRW: THE DEGENERATE BASE CASE

For spatially flat FLRW, $g = a(\eta)^2 \eta_{\rm Mink}$, the rail uses the conformal Killing vector ∂_η; the local speed is

$$v(\eta) = \sqrt{1 - a(\eta)^2/\hat{E}^2}, \tag{5}$$

vanishing at the freezing barrier $a = \hat{E}$.

a. Branch degeneracy. Because $a = a(\eta)$ depends on conformal time alone (spatial homogeneity), the optical index built from the conformal Killing vector,

$$n(\eta, r) = \frac{1}{v} = \frac{\hat{E}}{\sqrt{\hat{E}^2 - a(\eta)^2}}, \tag{6}$$

is *position-independent* at fixed η. The conformal-time functional $T_\eta = \int n \, d\ell_{\rm flat}$ is then minimized by a flat straight line (Fermat with a purely time-dependent index). The same curve minimizes T_t and T_τ: with $dt = a\, d\eta$ and $d\tau = a\sqrt{1 - \dot{\vec{x}}^2}\, d\eta$, the monotone reparametrization $t(\eta) = \int a\, d\eta$ maps critical points to critical points, and the τ-cost differs from the η-cost only by the same η-dependent weight. Hence

$$\begin{aligned} \arg\min T_t = \arg\min T_\tau &= \arg\min T_\eta \\ &= \text{comoving straight line.} \end{aligned} \tag{7}$$

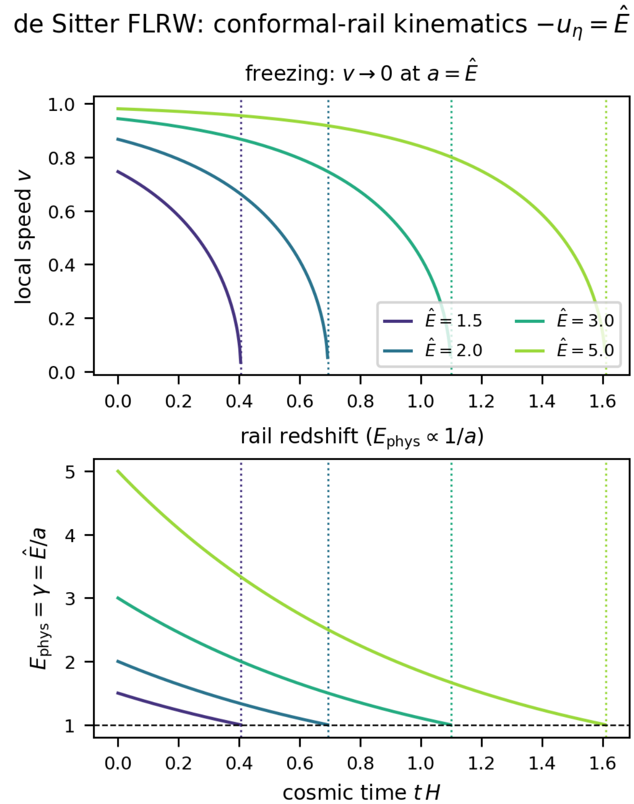

FIG. 2. de Sitter FLRW rail kinematics: local speed v and rail redshift $E_{\rm phys} = \hat{E}/a$; freezing at $a = \hat{E}$.

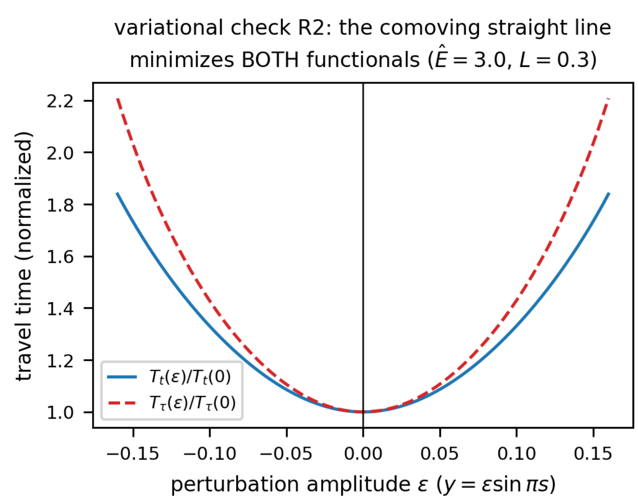

FIG. 3. Variational check: the comoving straight line minimizes *both* T_t and T_τ (branch degeneracy).

The t/τ splitting that drives every later result is a gravitational-*gradient* effect ($\partial_r f \neq 0$); it is absent whenever the metric functions depend on time only. FLRW is thus the degenerate base of the ladder: it exhibits freezing at $a = \hat{E}$ (Fig. 2) but no branch structure (Fig. 3).

IV. VAIDYA: DYNAMICAL BLACK HOLE AND THE KODAMA RAIL

The ingoing Vaidya metric $\mathrm{d}s^2 = -f\,\mathrm{d}v^2 + 2\,\mathrm{d}v\,\mathrm{d}r + r^2\mathrm{d}\Omega^2$, $f = 1 - 2m(v)/r$ [11, 18], has no timelike Killing vector; its trapping structure is that of a dynamical horizon [19–21]. The Kodama vector [16, 17, 22]

$$K^a = \varepsilon^{ab}\nabla_b r = \partial_v, \qquad \nabla_a K^a = 0, \tag{8}$$

is divergence-free without any Killing symmetry, and its charge is the Misner–Sharp mass $m(v)$ [23, 24].

A. Kodama energy is the rail invariant

From the rail constraint alone one finds the exact identity $-g(T,T) = (f\,u^v - 1)^2/\hat{E}^2$, hence

$$-\,u\cdot K \;=\; \hat{E} \quad \text{identically along the brachistochrone.} \tag{9}$$

This is *controlled* conservation, not Noether: for a geodesic $-u\cdot K$ would drift at rate $-m'(u^v)^2/r \propto m'$, exactly the right-hand side of Eq. (4) for $W = K$. Only the unnormalized $K = \partial_v$ has zero control cost in the static limit; the normalized $\hat{K} = K/\sqrt{f}$ does not (discriminating test), which is why the Kodama vector, not a unit selector, is the canonical choice.

a. Indicatrix and Hamiltonians. With $w \equiv \hat{E}^2 - f$ the constraint $-u_v = \hat{E}$ turns the admissible velocities at fixed (v, r) into an ellipse,

$$r' = (f - \hat{E}^2) + \hat{E}\sqrt{w}\,\cos\theta, \qquad \varphi' = \frac{\sqrt{w}}{r}\sin\theta, \tag{10}$$

a Zermelo problem with ingoing radial wind $f - \hat{E}^2 < 0$. Maximizing $p\cdot\dot{x}$ over θ at fixed $p_\varphi = J$ gives the closed-form branch Hamiltonians

$$H_v = p_r(f - \hat{E}^2) - 1 + \sqrt{w}\,\sqrt{\hat{E}^2 p_r^2 + J^2/r^2}, \tag{11}$$

$$H_\tau = p_r(f - \hat{E}^2) - \hat{E} + \sqrt{w}\,\sqrt{(\hat{E}p_r + 1)^2 + J^2/r^2}. \tag{12}$$

The τ-branch is the v-branch under $p_r \to p_r + 1/\hat{E}$ with unit cost replaced by $\hat{E}$. Non-stationarity is the explicit dependence $\mathrm{d}H/\mathrm{d}v = \partial_v H \propto m'(v)$; the teleological momentum $p_v = -H$ satisfies $p_v(r_1) = 0$ at the free arrival radius yet $p_v(r) \neq 0$ along the path, so the turning point *anticipates future accretion* (Fig. 5).

B. Phenomenology: penetration, timing, bounce

Three features distinguish the dynamical hole from its frozen-mass Schwarzschild limit.

Penetration. The τ-brachistochrone reaches the apparent horizon $r = 2m(v)$ only if $|J|$ lies below a threshold $J_c(v_0)$; accretion ($m' > 0$) opens a *finite* window

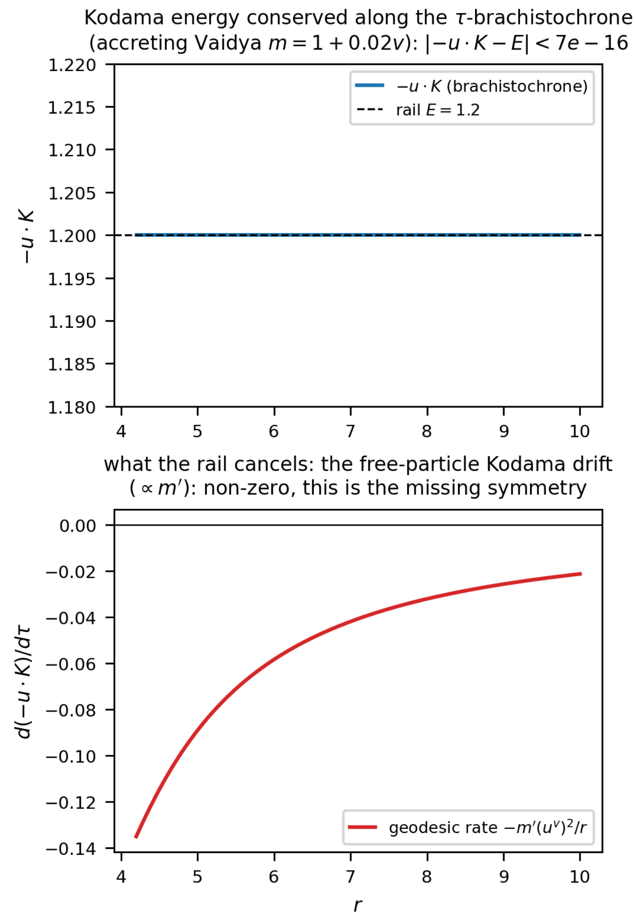

FIG. 4. Kodama energy $-u\cdot K = \hat{E}$ conserved along the τ-brachistochrone to 6.7×10^{-16} (top); the free-particle drift $\propto m'$ that the rail cancels (bottom).

$J_c > 0$, whereas evaporation ($m' < 0$) closes it to zero measure—an accretion/evaporation asymmetry with no static analogue. The finite capture interval $J \in (0, J_c]$ is the analogue of the Kerr t-branch dichotomy, generated here by the *dynamics*: grazing orbits linger near periapsis while the growing horizon closes the gap.

Bounce. The r-parametrization degenerates at periapsis ($\mathrm{d}v/\mathrm{d}r \to \infty$); reparametrizing by v makes the Euler–Lagrange problem regular at the turning point $r' = 0$, so the trajectory is integrated straight *through* the bounce—the same extremal, via the exact change of variable $(fv' - 1)\mathrm{d}r = (f - r')\mathrm{d}v$. The v-flow reaches the periapsis exactly where the r-flow diverges ($r_{\min} = 3.0105$ at $v = 7.29$ for $\mu \equiv m' = 0.01$), and the teleological memory $p_v(r) \propto m'$ measures the departure from the frozen-mass Schwarzschild curve (Fig. 20).

Timing and the orbit–horizon race. Because m grows along the path, $r_{\min}$ depends on *when* the particle arrives, the epoch v_1—a genuinely non-stationary effect absent in Kerr. Strikingly, the absolute and relative penetration have *opposite* winners (Fig. 21): under evaporation the absolute periapsis descends but the horizon $2m(v)$ retreats faster, so the relative penetration $r_{\rm min}/2m(v_{\rm peri})$ *worsens* for later arrivals; under accretion the periapsis rises yet the horizon grows faster, so late arrivals graze the relative horizon. The orbit–horizon chase has reversed outcomes measured in r versus in horizon units.

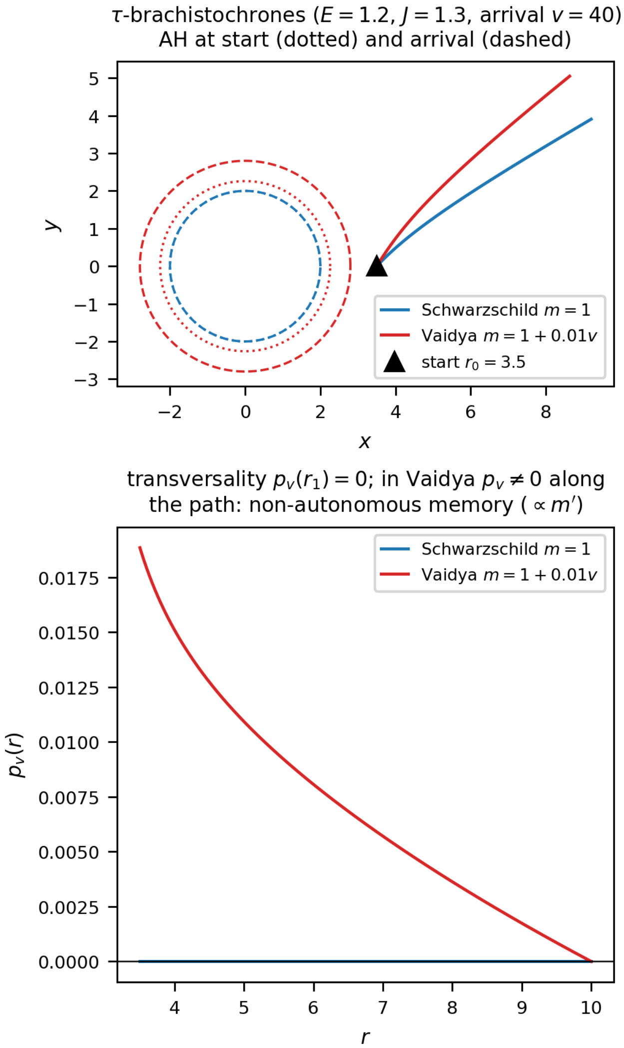

FIG. 5. τ-brachistochrones Schwarzschild vs Vaidya, and $p_v(r)$: transversality $p_v(r_1) = 0$, non-autonomous memory $\propto m'$.

C. Plunge-radius law and the (absent) evaporative inversion

The turning-point law has the static Schwarzschild form evaluated at the mass at periapsis, $f_* = 1 - 2m(v_{\rm peri})/r_{\rm min}$:

$$\tau: \quad (\hat{E}^2 - f_*)J^2 = f_* r_{\rm min}^2, \tag{13}$$

$$t: \quad (\hat{E}^2 - f_*)J^2 = \hat{E}^2 r_{\rm min}^2/f_*, \tag{14}$$

plus a genuine $O(m')$ correction from the teleological momentum p_v. The static ratio $J_t^2/J_\tau^2 = \hat{E}^2/f^2 > 1$ gives $r^t_{\rm min} < r^\tau_{\rm min}$ ("t deeper"), robust for physical evaporation: *there is no physical evaporative inversion* (the apparent linear-model inversion is an artifact of $m \le 0$).

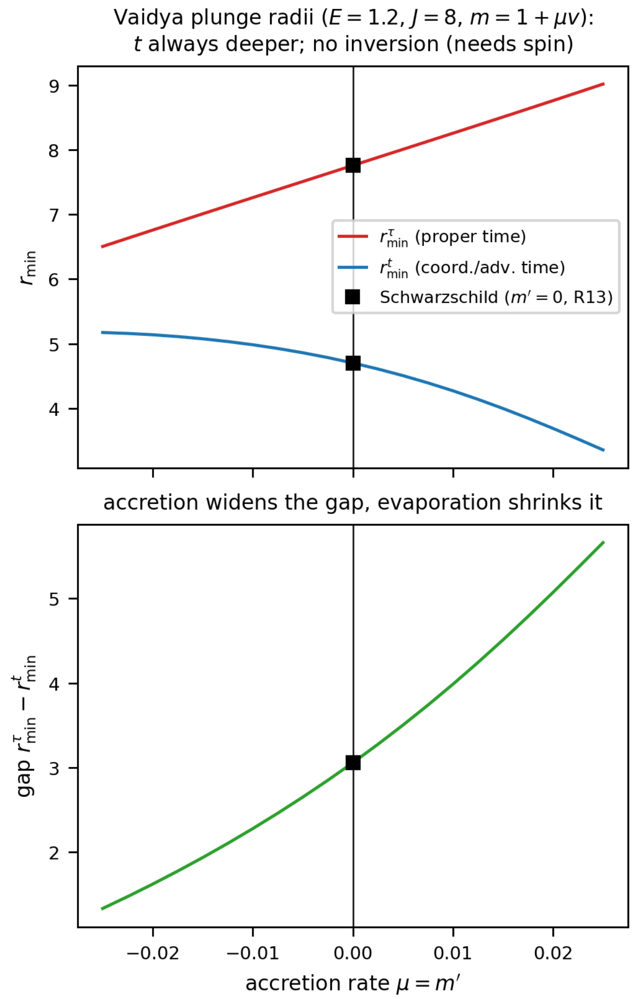

FIG. 6. Plunge radii $r^{t,\tau}_{\rm min}$ and gap vs $\mu = m'$; t deeper, accretion widens the gap.

V. THAKURTA–KERR: CONFORMAL ROTATING BLACK HOLE

Thakurta–Kerr is conformal Kerr, $g = A(\eta)^2 g_{\rm Kerr}$ [12, 25], one of a family of cosmological black holes [21, 26–30]. For constant A the transfer theorem maps the problem to Kerr with $E_{\rm eff} = \hat{E}/A$, $J_{\rm eff} = J/A$; the horizon r_+ is rigid (conformally invariant), the ergosphere $r_e = 2M$ is crossed regularly, while the freezing surface $A^2 f = \hat{E}^2$ breathes.

a. Indicatrix and Hamiltonians. Equatorially, write $\Delta = r^2 - 2Mr + s^2$, $\bar{v}^2 = 1 - A^2 f/\hat{E}^2$, and $\bar{P} =$

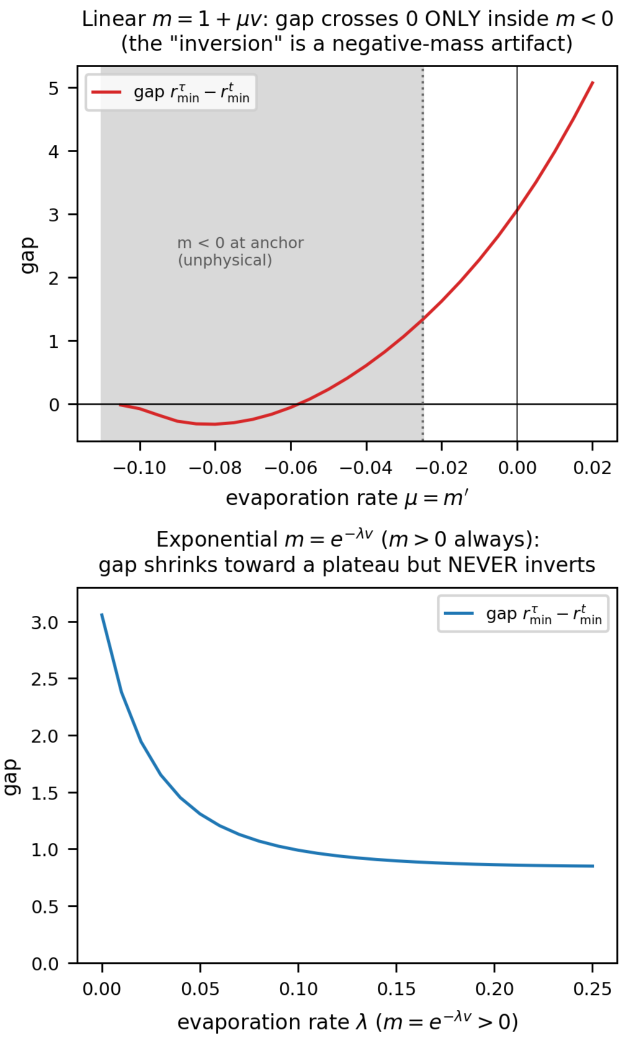

FIG. 7. No physical evaporative inversion: the linear-model sign change lies entirely in the unphysical $m < 0$ region; with $m = e^{-\lambda v} > 0$ the gap shrinks but never inverts.

$P + A^2(2Ms/r)^2/\hat{E}^2$ with $P = r^2 + s^2 + 2Ms^2/r$. The rail $\hat{E} = -u_\eta$ makes the indicatrix an ellipse with an *azimuthal* wind (frame dragging),

$$\varphi_0' = \frac{2Ms}{r}\frac{\bar{v}^2}{\bar{P}}, \qquad R^2 = f\bar{v}^2 + \bar{P}\varphi_0'^2 = \frac{\Delta\,\bar{v}^2}{\bar{P}}, \tag{15}$$

the last equality using the Kerr identity $fP + (2Ms/r)^2 = \Delta$. The conformal-time and proper-time Hamiltonians are

$$H_\eta = p_\varphi \varphi_0' + R\sqrt{(\Delta/r^2)p_r^2 + p_\varphi^2/\bar{P}} - 1, \tag{16}$$

$$H_\tau = \tilde{p}_\varphi \varphi_0' + R\sqrt{(\Delta/r^2)p_r^2 + \tilde{p}_\varphi^2/\bar{P}} - \frac{A^2 f}{\hat{E}}, \tag{17}$$

with the conformal gravitomagnetic shift $\tilde{p}_\varphi = p_\varphi - 2MsA^2/(r\hat{E})$. Both are regular through the ergosphere ($f = 0 \Rightarrow R^2 = \bar{v}^2\Delta/\bar{P} > 0$) and degenerate only at the horizon $\Delta = 0$, which is therefore conformally invariant—independent of A and $\hat{E}$. The three branches share this ellipse: t follows the same curves as η through $t = \int A\,\mathrm{d}\eta$, while τ is Riemannian-pure, the wind cancelling the gravitomagnetic term, so it survives the expansion (Fig. 8).

b. Non-autonomous optical reduction. Solving the indicatrix for $\mathrm{d}\eta$ at fixed spatial displacement casts the η-branch as a closed-form Randers metric, non-autonomous through the conformal factor,

$$\mathrm{d}\eta = n(\eta, r)\,\alpha_K + \beta_K, \qquad n = \frac{\hat{E}}{\sqrt{\hat{E}^2 - A^2 f}}, \tag{18}$$

$$\alpha_K^2 = \frac{r^2\,\mathrm{d}r^2}{f\Delta} + \frac{\Delta\,\mathrm{d}\varphi^2}{f^2}, \qquad \beta_K = -\frac{2Ms}{rf}\,\mathrm{d}\varphi. \tag{19}$$

The Riemannian data α_K and the one-form β_K are *exactly* those of the *null* Fermat (Randers) metric of Kerr [5]: all of the mass, rail, and expansion physics is carried by the scalar index n, which multiplies only the Riemannian part, while the frame-dragging one-form β_K is rigid—conformally invariant and independent of $\hat{E}$. The reduction is thus a Zermelo problem with a time-dependent landscape in which only the metric term breathes, not the wind. It degenerates at the ergosphere ($f = 0$, where $1/F$ is intrinsic, as for stationary Kerr) even though the Hamiltonian (16) remains regular there; the limits $\hat{E} \to \infty$ (null Kerr optics), $A = 1$ (stationary Perlick on Kerr), $s = 0$ (Thakurta–Schwarzschild) and $M = s = 0$ ($n = 1/v$, FLRW) are all recovered.

A. Quasi-constants and drift

For constant A the transfer theorem is exact at the level of conserved quantities: $u_{\mathrm{TK}} = u_{\mathrm{K}}/A$ maps the $\hat{E}$-rail onto the Kerr rail with $E_{\mathrm{eff}} = \hat{E}/A$, so every Kerr quasi-constant carries over by substitution,

$$K^{\mathrm{TK}} = K^{\mathrm{Kerr}}(E_{\mathrm{eff}}), \qquad K_\tau = Q_{\mathrm{std}} + a^2 f_2\, p_\theta^2 + O(a^4), \tag{20}$$

with Q_{std} the Carter constant and f_2 the next-to-leading polar coefficient [8, 31]. The polar libration that carries the quasi-constant lives off the equatorial plane, governed by the full Hamiltonian on a generic slice $\Sigma = r^2 + s^2\cos^2\theta$,

$$H_\tau = \tilde{p}_\varphi \varphi_0' + R\sqrt{\frac{\Delta}{\Sigma}p_r^2 + \frac{p_\theta^2}{\Sigma} + \frac{\tilde{p}_\varphi^2}{\bar{G}}} - \frac{A^2 f_\Sigma}{\hat{E}}, \tag{21}$$

with $f_\Sigma = 1 - 2Mr/\Sigma$, $b = 2Mar\sin^2\theta/\Sigma$, $\bar{G} = G + A^2b^2/\hat{E}^2$, $\tilde{p}_\varphi = J - A^2b/\hat{E}$, $\varphi_0' = b\bar{v}_\Sigma^2/\bar{G}$, $R^2 = \bar{v}_\Sigma^2\Delta\sin^2\theta/\bar{G}$, and the block identity $f_\Sigma G + b^2 = \Delta\sin^2\theta$; it reduces to Eqs. (16)–(17) on the equator. For $A = A(\eta)$ the substitution (20) is only the zeroth order: the transport equation $\mathrm{d}K/\mathrm{d}\lambda = \{K, H\} + \partial_\eta K$ acquires an ex-

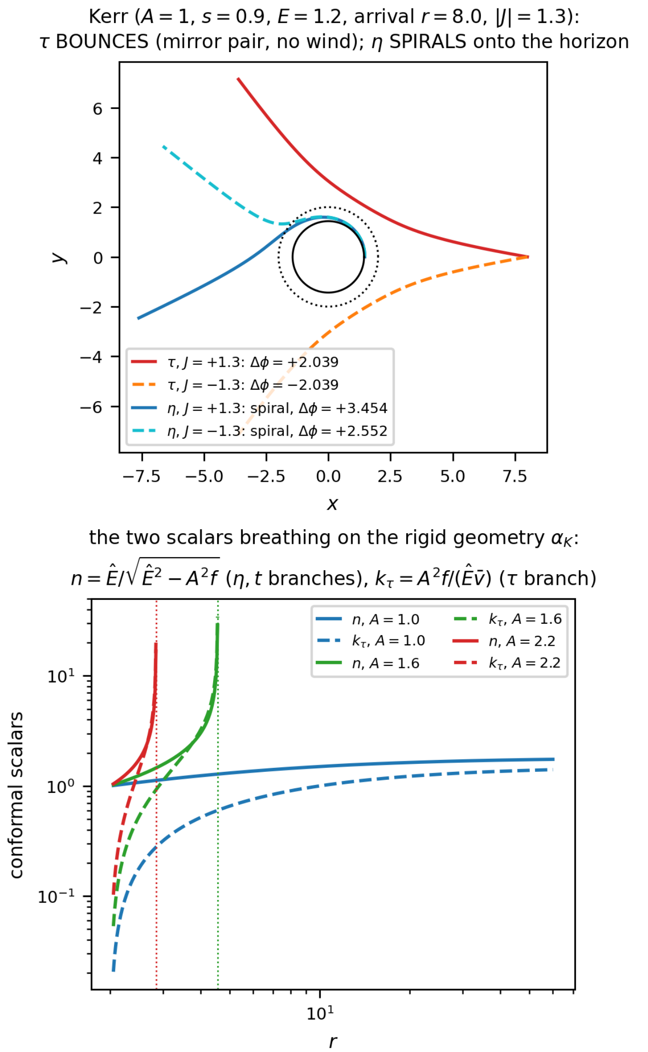

FIG. 8. The three branches: τ bounces (mirror pair, no wind), η spirals (gravitomagnetic wind); breathing conformal scalars n, k_τ.

plicit source at order $a^2\varepsilon$, $\varepsilon = A'/A$,

$$\left.\frac{\mathrm{d}K}{\mathrm{d}\eta}\right|_{\text{expl}} = \varepsilon\, a^2\, S_1, \qquad S_1 = 2E_{\text{eff}}^2\left[\cos^2\theta + \frac{M\cos 2\theta\, p_\theta^2}{r\, DE_0^2}\right], \tag{22}$$

where $DE_0 = r^2 w_{\text{eff}}$. The counterterm h solving $\{h, H_0\} = -S_1$ inherits the double pole at the *freezing wall* $r_w = 2M/(1-E_{\text{eff}}^2)$, so the dynamical quasi-constant is

$$K^{\text{TK}}(\eta) = K^{\text{Kerr}}\big(E_{\text{eff}}(\eta)\big) - a^2\int \varepsilon\, S_1\, \mathrm{d}\eta, \tag{23}$$

with closed toroidal averages $\langle S_1\rangle = E_{\text{eff}}^2(1 - J^2/L^2) + 2ME_{\text{eff}}^2|J|(L-|J|)/(r\,DE_0^2)$, $L^2 = p_\theta^2 + J^2/\sin^2\theta$. In Kerr ($E > 1$) the wall is absent; in Thakurta–Kerr with $E_{\text{eff}} < 1$ it enters from the far field and advances inward as A grows, becoming a *global attractor*: the bound τ-libration ceases to exist, so the dynamical quasi-constants live on scattering arcs ($E_{\text{eff}} > 1$) or on the t-branch.

VI. EQUATORIAL CLOSED FORMS AND THE ERGOSPHERE

A. Separatrix in Weierstrass functions

From the doranTau factorization $\Delta - J_c^2 w = f(r^2 + c^2)$, $c = a/E$ [8], the equatorial τ-separatrix $\varphi(r)$ is genus 1, closed in the Weierstrass functions $\wp, \sigma, \zeta$ [32–35]: the angle is linear in the uniformizing variable $z(r) = \wp^{-1}(A/r+B)$ plus two σ-ratios [App. A 1, Eq. (A3)], regular *through* the ergosphere because the factor $r^2 + c^2$ divides the turning quartic and removes r_e from the radicand. The Boyer–Lindquist form winds logarithmically at the horizon; the Doran shift—a second, quasi-lemniscatic elliptic curve—cancels it, so the ergosphere-penetrating separatrix $\varphi_D = \varphi_{\text{BL}} + \varphi_{\text{shift}}$ is closed on two elliptic curves. The Kerr geodesic structure and its constants are classical [31, 36]. Validated end-to-end to 2.8×10^{-10} through the ergosphere, and general across parameters.

B. Trichotomy and the ergosphere cusp

For the equatorial τ-branch, $\mathrm{d}\varphi/\mathrm{d}r \sim K\sqrt{r - r_e}$ as $r \to r_e$ (because $\sqrt{f} \to 0$ while the turning radicand stays finite), giving

$$\varphi - \varphi_e \sim \tfrac{2}{3}K\,(r - r_e)^{3/2}, \tag{24}$$

a cusp with radial tangent. The trichotomy in $|J|$ (sign of $a^2 - J^2E^2$): smooth periapsis above r_e ($|J| > J_c$); smooth crossing at r_e ($|J| = J_c = a/E$, separatrix); cusp reflection ($|J| < J_c$). The crossing branch is continued through r_e to the horizon in Doran (horizon-penetrating) coordinates [37–39], $\varphi_D = \varphi_{\text{BL}} - \varphi_{\text{shift}}$.

a. Retrograde branch and the τ/t asymmetry. The trichotomy above is a τ-branch statement, and it is *symmetric* in the sign of J: because the doranTau factorization depends on J only through J^2, the τ-capture band—the momenta for which the ingoing orbit reaches r_e—is $|J| \le J_c = a/E$, prograde and retrograde sharing the same threshold despite frame dragging. The t-branch is not protected. Its turning function (from $H_\eta = 0$ at $p_r = 0$, with $b = 2Ma/r$) is

$$N(r) = \left(\bar{P} - Jb\bar{v}^2\right)^2 - J^2\Delta\bar{v}^2, \tag{25}$$

which is only linear-plus-quadratic in J. At the ergosphere $N(r_e) = \bar{P}_e\,(\bar{P}_e - 2aJ)$ with $\bar{P}_e = \bar{P}(r_e) = 4M^2 + 2a^2 + a^2/E^2$, so the t-orbit crosses r_e at the single *prograde* value

$$J_+^t = \frac{\bar{P}_e}{2a} = \frac{4M^2 + 2a^2 + a^2/E^2}{2a}; \tag{26}$$

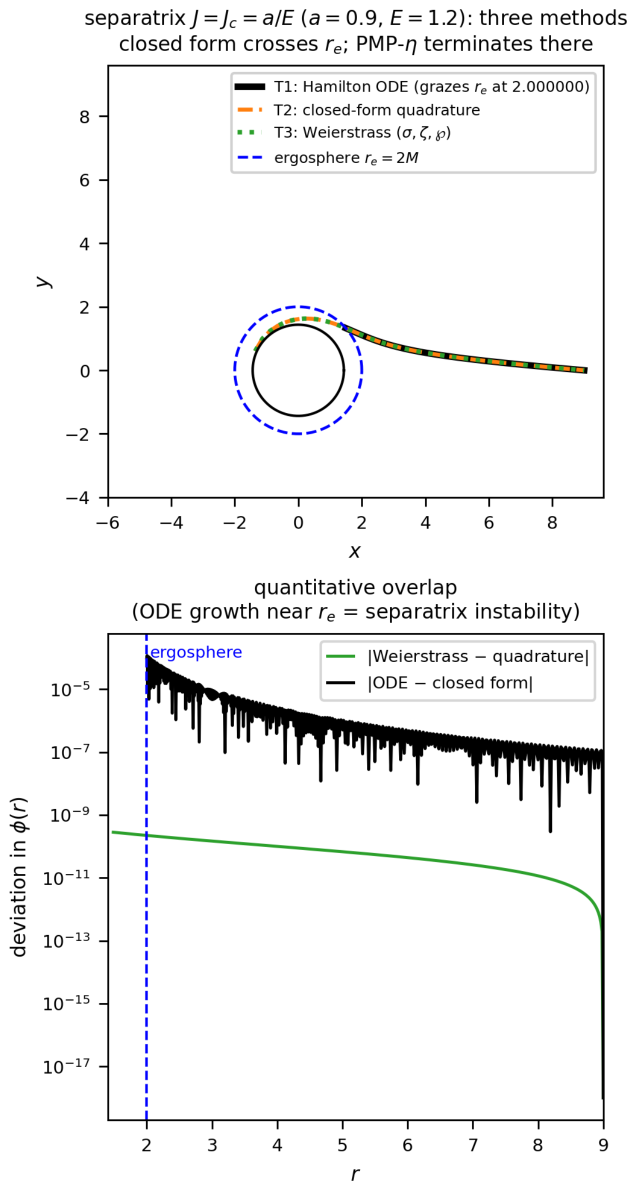

FIG. 9. Separatrix $J = J_c$: ODE, quadrature and evaluated Weierstrass agree to 2.8×10^{-10}; the PMP-η terminates at r_e.

there is no retrograde r_e-crossing. Instead the retrograde t-capture is bounded by the *marginal* (double-root) condition $N = N' = 0$ at a radius $r_* > r_e$—the massive analogue of the retrograde photon sphere—which closes to

$$J_c^- = -\left.\frac{\bar{P}}{\bar{v}\,(\sqrt{\Delta} - b\bar{v})}\right|_{r_*} < 0. \tag{27}$$

For $a = 0.9$, $E = 1.2$ (and $A = 1$) these give $J_+^t = 3.43$ and $J_c^- = -8.05$ ($r_* = 3.51$): the t-branch captures the wide, asymmetric band $J \in [J_c^-, J_+^t]$, far more capture-prone on the retrograde side. A moderately retrograde orbit ($J = -5$, say) is therefore captured on the t-branch but scattered on the τ-branch. For constant A the results carry over by $E \to \hat{E}/A$, $a \to s$, $J \to J/A$; in particular $J_c^\tau = sA^2/\hat{E}$.

C. Plunge inversion: rotational and conformal

Two mechanisms invert the t/τ ordering. Rotationally, in the (a, J) plane, $\rho(r_*)^2 = H_{t0}/H_{\tau 0}$, extending the stationary Kerr analysis [8]. Conformally, the inversion locus is

$$\mathrm{cond}(r, A) = A^2 b\,Q - \bar{P}(\hat{E} - A^2 f) = 0, \quad Q = b\bar{v}^2 + \sqrt{\bar{v}^2\Delta}, \tag{28}$$

with $\mathrm{cond}(1) < 0$ and $\mathrm{cond}(A_{\mathrm{freeze}}) = \hat{E}r\Delta(\hat{E} - 1)/(r - 2M) > 0$, so by the intermediate value theorem $A_{\mathrm{inv}} \in (1, A_{\mathrm{freeze}})$: the conformal inversion always exists and is reachable (far field $A_{\mathrm{inv}} \to \sqrt{\hat{E}}$).

D. Fixed-endpoint comparison and protocol dependence

The statement "branch X plunges deeper" is protocol dependent. With *fixed initial data* (same r_0, J, p_r) the t-branch is deeper ($r^t_{\mathrm{min}} < r^\tau_{\mathrm{min}}$). With *fixed endpoints* (the genuine two-point brachistochrone), the two branches join the same A, B but are distinct paths, and the ordering reverses: $r^\tau_{\mathrm{min}} < r^t_{\mathrm{min}}$. The optical-index ratio $n_t/n_\tau = E/f > 1$ penalizes deep radii more heavily for the t-branch, so the t-brachistochrone bows outward while the τ-branch dips. This must be specified when comparing depths.

Crucially, neither driver of same-launch inversion survives the fixed-endpoint constraint. Scanning the black-hole spin over the entire Kerr flow with fixed symmetric endpoints $(r_0, \pm\Phi)$ leaves $\Delta r = r^t_{\mathrm{min}} - r^\tau_{\mathrm{min}}$ strictly positive ($\Delta r \in [+0.45, +2.06]$ on $a \in [0.05, 0.95]$); the map is nearly flat in a, its gradient set almost entirely by the endpoint angle Φ (encoded by J). The conformal scan (via the transfer $E_{\mathrm{eff}} = \hat{E}/A$) is likewise everywhere positive ($\Delta r \in [+0.002, +3.711]$), as forced by the theorem $n_t/n_\tau = E/f > 1$. Table I summarizes

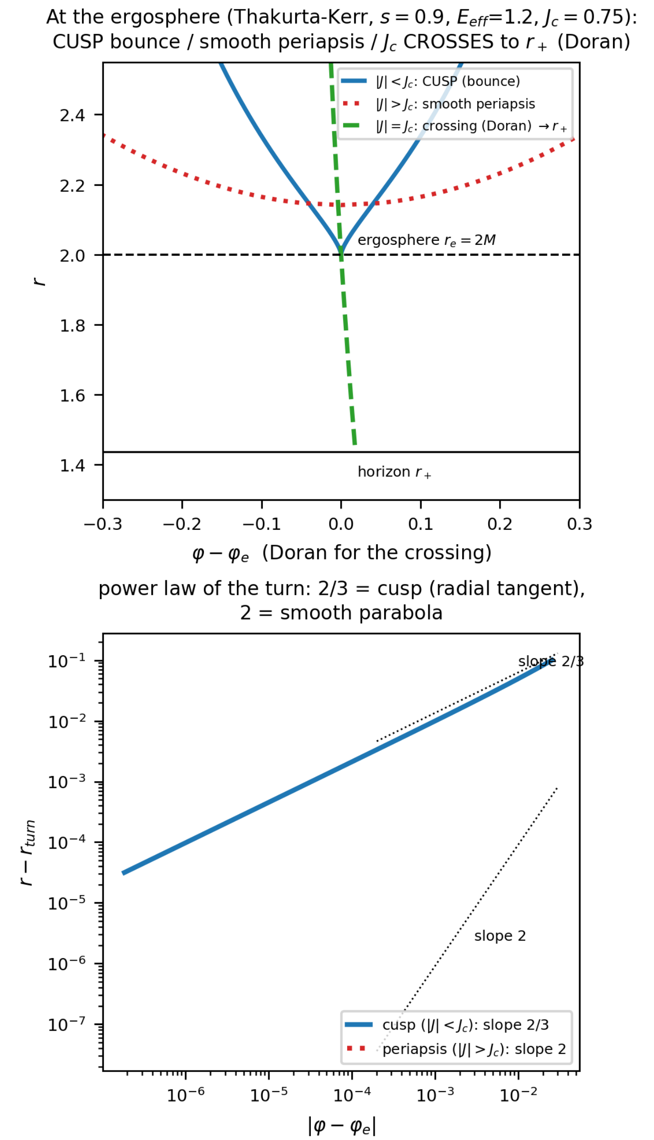

FIG. 10. Reflection at the ergosphere: cusp ($|J| < J_c$), smooth crossing to r_+ in Doran coordinates ($|J| = J_c$), smooth periapsis ($|J| > J_c$); power law $2/3$ vs 2.

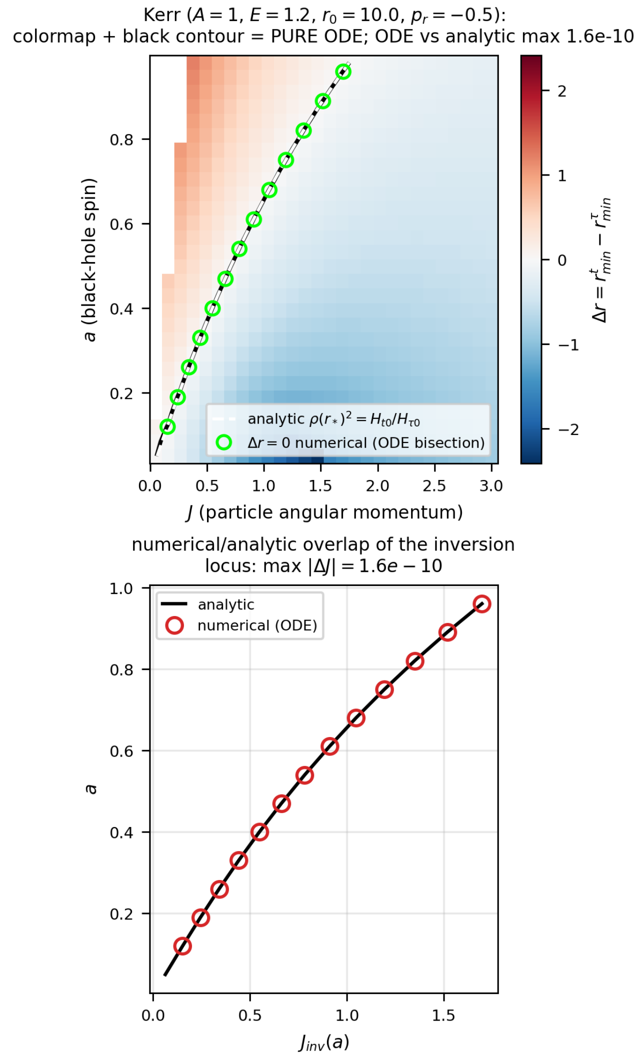

FIG. 11. Rotational inversion in the (a, J) plane; ODE locus vs analytic $\rho(r_*)^2 = H_{t0}/H_{\tau 0}$.

TABLE I. Protocol dependence of plunge inversion. Same-launch comparison inverts under either rotation or a conformal factor; the genuine fixed-endpoint brachistochrone never inverts.

protocol	spin a	conformal A
same launch	inverts (§VI C)	inverts (§VI C)
fixed endpoints	no ($\Delta r > 0$)	no ($n_t/n_\tau = E/f > 1$)

the resulting protocol dependence: inversion is a property of the *comparison protocol*, not an invariant of the two brachistochrones—it requires the freedom to let the endpoint move.

The verdict is independent of the endpoint symmetry: repeating both scans with *asymmetric* fixed endpoints ($r_A = 10, r_B = 6$) leaves $\Delta r > 0$ everywhere as well (spin $[+0.76, +2.64]$, conformal $[+0.16, +3.24]$; Figs. 16–17), consistent with the pointwise bound $n_t/n_\tau = E/f$.

VII. CONCLUSIONS

The non-stationary brachistochrone is a well-posed controlled-rail problem: the invariant $-u \cdot W = \hat{E}$ is legitimate via Pontryagin's principle, and the selector W is fixed canonically by the Killing–CKV–Kodama hierarchy, with the Kodama energy conserved along the rail even when free fall does not conserve it. FLRW is the de-

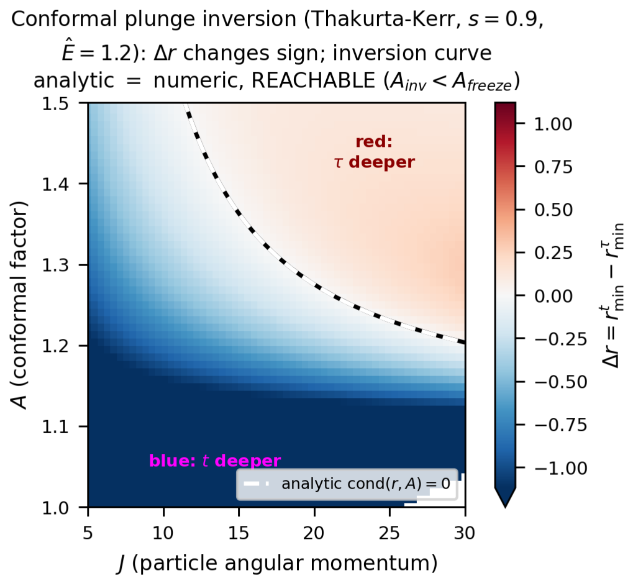

FIG. 12. Conformal inversion in the (J, A) plane: Δr changes sign; analytic cond $= 0$ matches the numerical contour; reachable ($A_{\mathrm{inv}} < A_{\mathrm{freeze}}$).

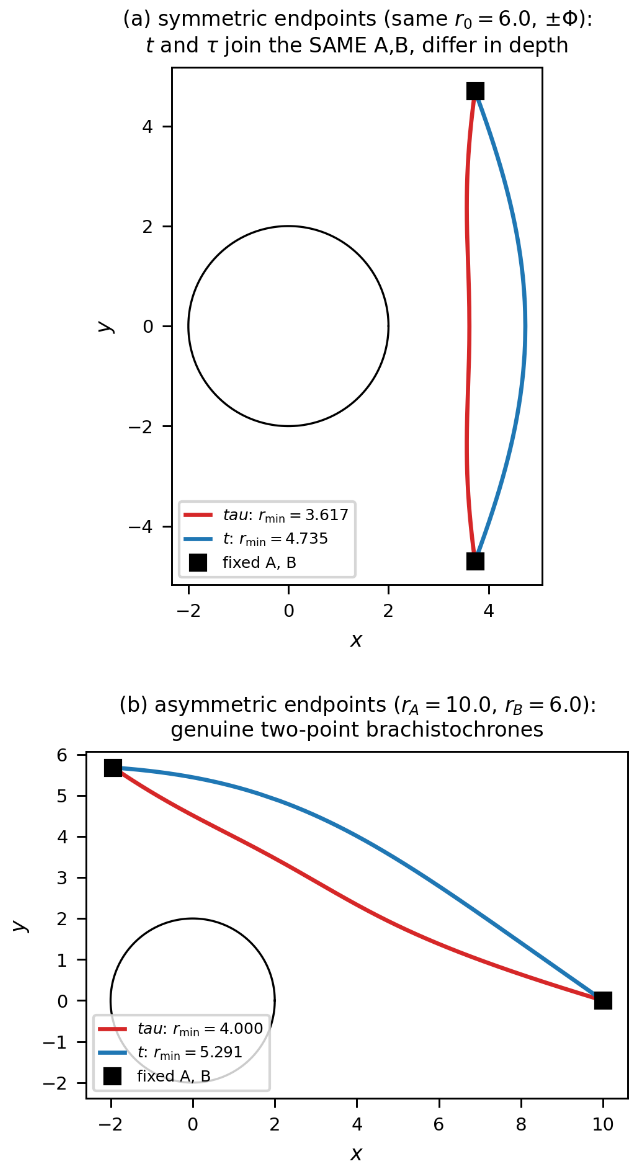

FIG. 13. Fixed-endpoint t and τ brachistochrones (Schwarzschild, $E = 1.2$): symmetric (top) and asymmetric (bottom) endpoints. Both branches join the same A, B; the τ-branch dips deeper.

generate base—freezing without branch splitting—while the black hole breaks the t/τ symmetry through the gravitational gradient. On the equatorial sector we obtained closed-form separatrices in Weierstrass functions, continued through the ergosphere in Doran coordinates, and an analytic cusp theorem for the $|J| < J_c$ reflection. The plunge inversion resolves into a three-way picture: rotation and the conformal factor invert the t/τ ordering under a same-launch comparison, whereas physical evaporation never does, and *no* inversion survives once both endpoints are fixed. Outlook: the exact Sultana–Dyer source [27], the McVittie/Schwarzschild–de Sitter limits [26, 28, 40] with their photon surfaces [41], escape maps, and freezing-surface dynamics.

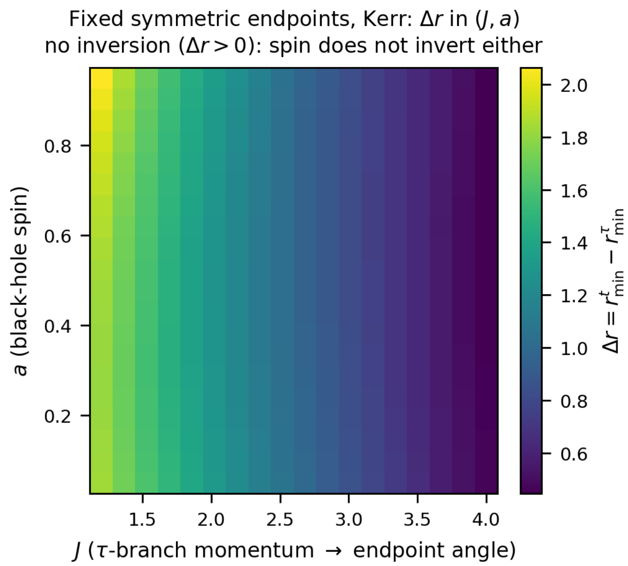

FIG. 14. Spin scan at fixed symmetric endpoints (Kerr, Hamiltonian flow): Δr in the (J, a) plane stays positive everywhere—spin does not invert once the endpoints are pinned. The gradient is set by the endpoint angle (J), not by a.

Appendix A: Validation and consistency checks

We collect the numerical checks that underpin the analytic results. The minimum principle is verified directly by perturbing the computed brachistochrones and confirming that both T_t and T_τ are minimized (Fig. 3 for the FLRW branch degeneracy, and Fig. 19 for the two distinct Vaidya branches). Stationary-limit residuals confirm that the non-stationary Hamiltonian flow reproduces the closed-form Kerr expressions as $m' \to 0$ and $A \to 1$ (Fig. 18); the Kodama energy is conserved to $\sim 10^{-16}$ along the rail (Fig. 4). The equatorial closed forms are validated end-to-end by the three-method superposition (ODE flow, quadrature, evaluated Weierstrass) through the ergosphere (Fig. 9), and the fixed-endpoint no-inversion theorem is checked over the asymmetric-endpoint grids (Figs. 16–17).

1. Explicit closed forms

a. Equatorial separatrix in Weierstrass functions. On the separatrix $J = J_c$ the doranTau factorization $\Delta - J_c^2 w = f(r^2 + c^2)$, $c = a/E$, makes the turning quartic $Q_4 = r\big((E^2-1)r + 2M\big)(r^2+c^2)$, and the factor $r^2 + c^2$ divides Q_4 so that only the horizon poles survive. Writing

$$
\begin{aligned}
A &= \frac{Ma^2}{2E^2}, \qquad B = \frac{a^2(E^2-1)}{12E^2},\\
g_2 &= \frac{a^4(E^2-1)^2}{12E^4} - \frac{a^2M^2}{E^2},\\
g_3 &= \frac{a^4\left[36E^2M^2(1-E^2) - a^2(E^2-1)^3\right]}{216E^6},
\end{aligned}
\tag{A1}
$$

the uniformizing variable is $z(r) = \wp^{-1}\big(A/r + B; g_2, g_3\big) = \int_0^r \mathrm{d}r'/\sqrt{Q_4}$, the Abel map of the turning quartic Q_4: it is the elliptic argument in which the radial motion linearizes, so that r is recovered by $A/r + B = \wp(z)$ and the physical range $r \in [r_{\min}, \infty)$ maps to a segment of z. With $r_\pm = M \pm \sqrt{M^2 - a^2}$ the horizons,

$$
\begin{aligned}
\alpha_\pm &= \frac{r_\pm\big((E^2-1)r_\pm + 2M\big)}{r_\pm - r_\mp},\\
\lambda_\pm &= \frac{\alpha_\pm}{\sqrt{Q_4(r_\pm)}}, \qquad v_\pm = z(r_\pm),
\end{aligned}
\tag{A2}
$$

$\wp(v_\pm) = A/r_\pm + B$, and $\Lambda_0 = (E^2-1) - \alpha_+/r_+ - \alpha_-/r_- + 2\lambda_+\zeta(v_+) + 2\lambda_-\zeta(v_-)$, the angle is linear in the uniformizing variable $z \equiv z(r)$ plus two Weierstrass-σ ratios,

$$
\begin{aligned}
\varphi(r) = \frac{a}{E}\Big[\Lambda_0\, z(r) &+ \lambda_+ \ln\frac{\sigma(z - v_+)}{\sigma(z + v_+)}\\
&+ \lambda_- \ln\frac{\sigma(z - v_-)}{\sigma(z + v_-)}\Big] + \text{const.}
\end{aligned}
\tag{A3}
$$

Because $Q_4(2M) = 4M^2(4E^2M^2 + a^2) > 0$, Eq. (A3) is regular *through* the ergosphere $r_e = 2M$; the only singularities are the third-kind logarithms at the horizons. The conformal case is $E \to E_{\text{eff}} = \hat{E}/A$ in every parameter.

b. Doran continuation through the horizon. The Boyer–Lindquist form $\varphi(r)$ winds logarithmically at r_+ (the standard BL artifact); adding the Doran shift $\mathrm{d}\varphi_{\text{shift}} = 2Mar\,\mathrm{d}r/(\Delta\sqrt{C_3})$ with the cubic $C_3 = 2Mr(r^2 + a^2)$ regularizes it. This is a *second* elliptic curve, in clean Weierstrass form $r(z) = (2/M)\wp(z; g_2, g_3)$ with $g_2 = -M^2a^2$, $g_3 = 0$ (quasi-lemniscatic), giving $\varphi_{\text{shift}} = \frac{a}{r_+ - r_-}\big(B_+ - B_-\big)$, $B_\pm = 2\zeta(v_\pm)z + \ln[\sigma(z - v_\pm)/\sigma(z + v_\pm)]$; the log divergences of φ and φ_{shift} cancel at r_+, the analytic content of Doran regularity. Thus $\varphi_D = \varphi_{\text{BL}} + \varphi_{\text{shift}}$ is closed on two elliptic curves.

c. Stationary-limit conserved quantities. At $A = 1$ the branch flows admit the closed first integrals used for validation,

$$
\mathcal{K}_\tau = J, \qquad \mathcal{K}_t = \frac{fJ + 2Ms/r}{E}, \qquad \mathcal{K}_\eta = \frac{fJ}{E}, \tag{A4}
$$

the equatorial Kerr doranTau/doranT constants.

d. Rotational inversion function. The same-launch t/τ ordering inverts where $\rho(r_*)^2 = H_{t0}/H_{\tau 0}$ with the explicit ratio of centrifugal potentials

$$
\rho = \frac{A^2}{\hat{E}}\Big(f + \frac{2Ms}{rJ}\Big), \tag{A5}
$$

$\rho < 1$ giving t deeper, $\rho > 1$ giving τ deeper.

e. Conformal trichotomy. With constant A the separatrix momentum is $J_c = sA^2/\hat{E}$, and above it the smooth periapsis is the outer root of

$$
\Delta(r) - \Big(\tfrac{J}{A}\Big)^2\Big[\Big(\tfrac{\hat{E}}{A}\Big)^2 - f(r)\Big] = 0, \tag{A6}
$$

while $|J| < J_c$ reflects at the cusp $r_{\min} = r_e$ and $|J| = J_c$ crosses to r_+.

f. Closed toroidal averages. The secular source $\langle S_1\rangle$ of Eq. (22) follows from the orbit-plane averages ($\cos\theta = \sin i\,\sin\psi$, $k = 1 - J^2/L^2$)

$$
\Big\langle \cos^2\theta \Big\rangle = \tfrac{k}{2}, \quad \Big\langle \tfrac{1}{\sin^2\theta}\Big\rangle = \tfrac{L}{|J|}, \quad \big\langle p_\theta^2\big\rangle = L(L - |J|), \tag{A7}
$$

with $\langle \cos 2\theta\, p_\theta^2\rangle = |J|(L - |J|)$.

2. Consistency residuals

The closed forms and reductions above were checked symbolically and numerically. The Weierstrass separatrix (A3) reproduces the direct quadrature to $|\Delta\varphi| < 2.8\times 10^{-10}$ through the ergosphere, with the $\wp$-identity satisfied to 9×10^{-28} and the antiderivative to 30 digits; the Doran shift reduces to 1.1×10^{-16}. The non-autonomous Randers data (19) match the Kerr null Fermat metric to $\sim 10^{-14}$. The equatorial Thakurta–Kerr flow reproduces the $A = 1$ Kerr constants (A4) to ten digits at $r = 4, 6, 9$ ($a = 0.9$, $E = 1.2$), and the $s \to 0$

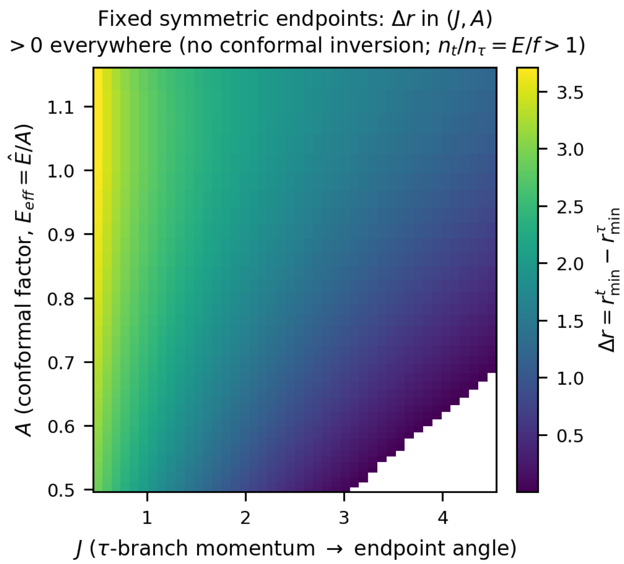

FIG. 15. Conformal scan at fixed symmetric endpoints (via $E_{\rm eff} = \hat{E}/A$): Δr in the (J, A) plane is positive everywhere, matching the closed-form theorem $n_t/n_\tau = E/f > 1$.

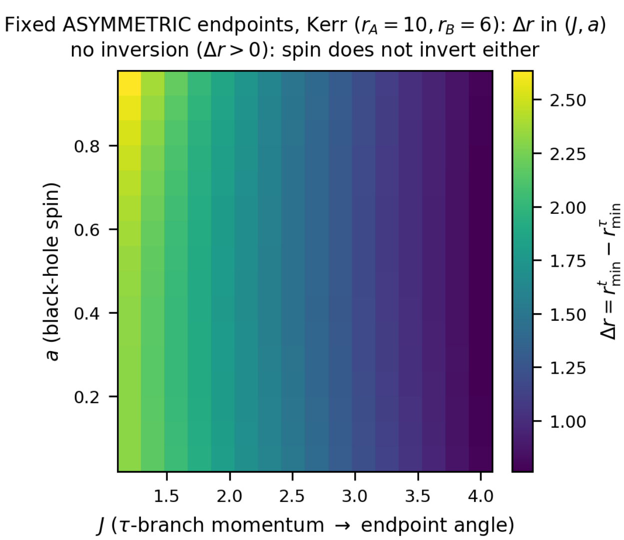

FIG. 16. Robustness of the fixed-endpoint no-inversion result under *asymmetric* endpoints ($r_A = 10, r_B = 6$), spin scan (Kerr flow): $\Delta r > 0$ throughout the (J, a) plane.

limit the Thakurta–Schwarzschild flow to 10^{-12}; the 3+1 Hamiltonian (21) reduces to the equatorial one exactly, and conserves L^2 to 6×10^{-13} at $a = 0$. The Kodama energy is held to $|-u \cdot K - \hat{E}| < 7 \times 10^{-16}$ along the rail, and the static inversion/trichotomy colormap agrees with the analytic contours (A6) to a relative 7×10^{-12}.

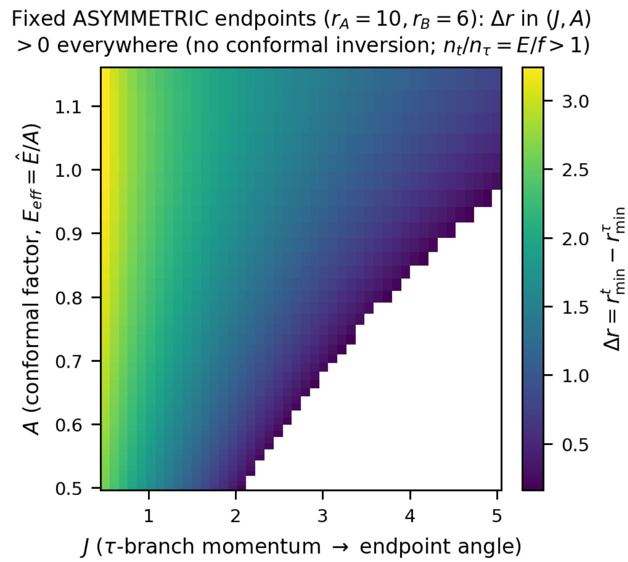

FIG. 17. Same robustness check, conformal scan (via $E_{\rm eff} = \hat{E}/A$): $\Delta r > 0$ throughout the (J, A) plane for asymmetric endpoints. The blank wedge is the region where the τ-turning point rises above r_B (no dipping orbit reaches B).

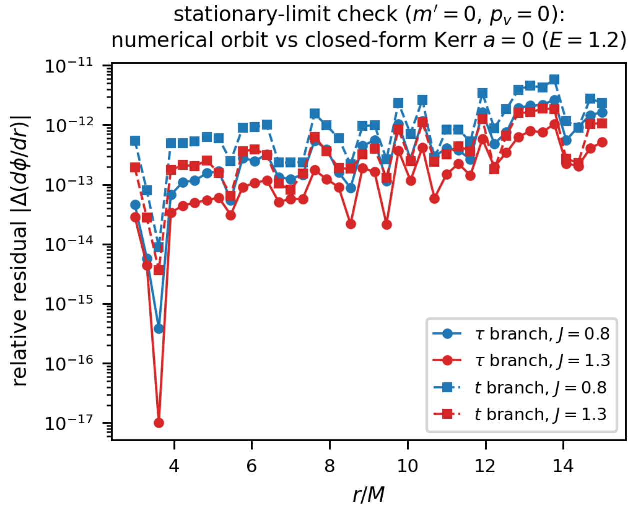

FIG. 18. Stationary-limit check ($m' = 0$): numerical orbit vs closed-form Kerr $a = 0$.

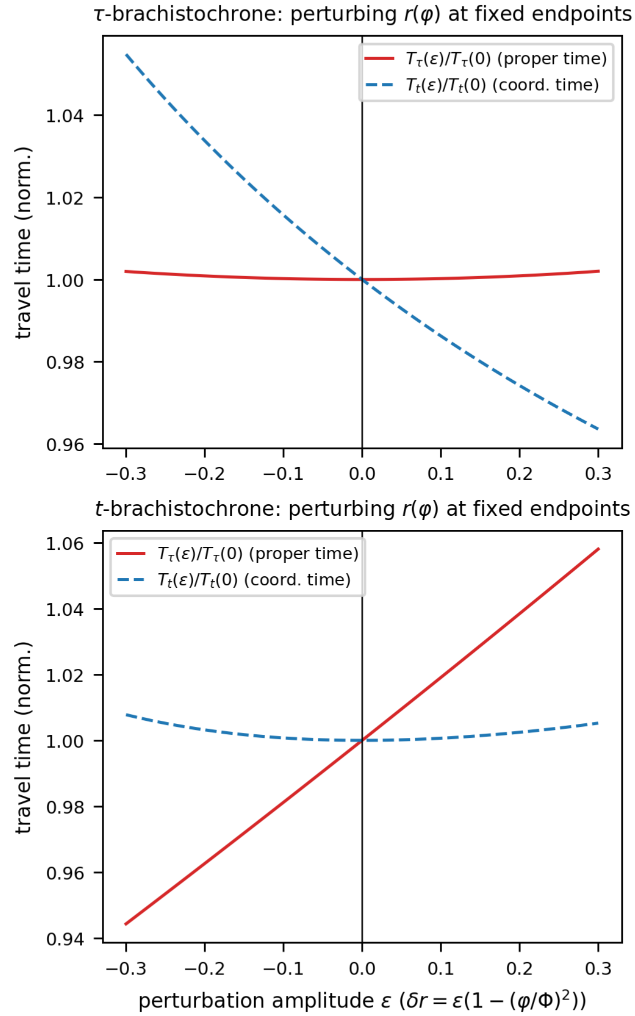

FIG. 19. Each branch minimizes its own travel time; t and τ are distinct curves.

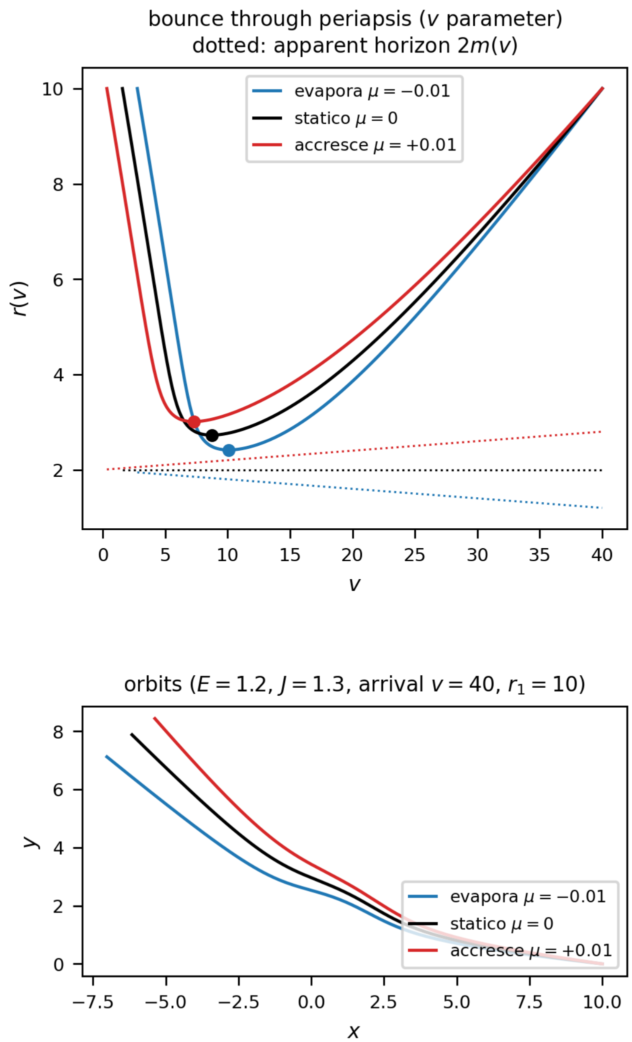

FIG. 20. Bounce through periapsis in the v-parametrization: the Euler–Lagrange flow is regular at $r' = 0$ and integrates through the turning point, reaching $r_{\min}$ exactly where the r-parametrization diverges.

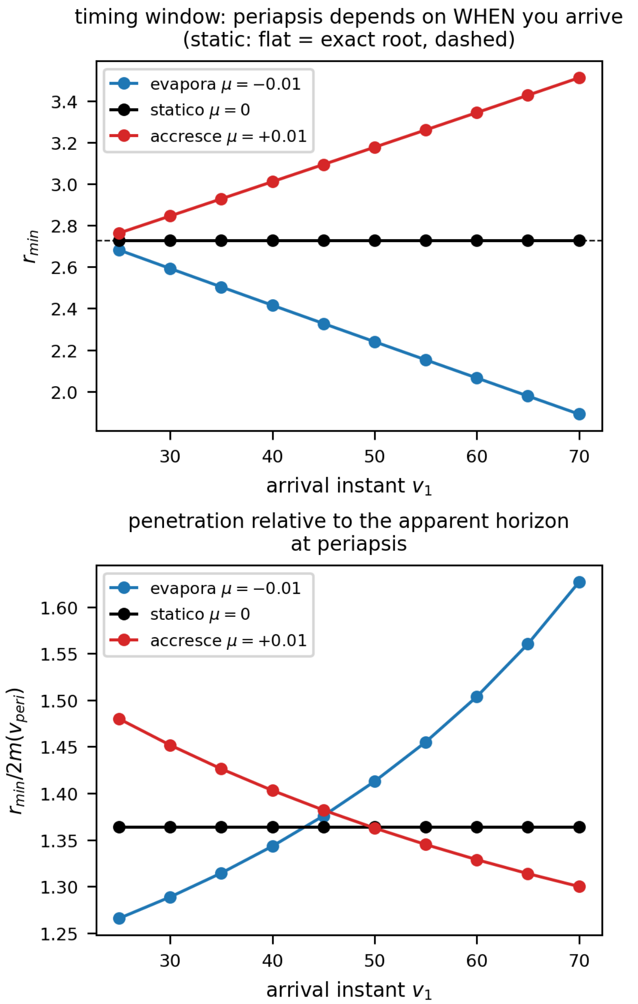

FIG. 21. Timing window: $r_{\min}$ depends on the arrival epoch v_1. The orbit–horizon race has opposite winners in absolute ($r_{\min}$) and relative ($r_{\min}/2m$) terms for accretion vs evaporation.

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DATA AVAILABILITY

The complete numerical and symbolic codebase underlying this work is openly available under the MIT license in the GitHub repository "NonStationaryBrachistochrone" [42]:

https://github.com/ImanetorX98/NonStationaryBrachistochrone

The repository includes the trajectory integrators, the Pontryagin Hamiltonian constructions, the Poisson-bracket closure and quasi-invariant scans (Sec. V A), the Weierstrass separatrix and Doran-continuation scripts, the fixed-endpoint BVP and colormap pipelines, and the figure-generation code. It also retains exploratory scripts and negative results, documenting which ansatz classes failed the held-out trajectory and phase-space residual tests reported in Sec. V A and App. A.

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